

GRIDGUARD AI

Predictive Power Grid Resilience &
Crew Pre-positioning Platform

Team GridGuard AI



Power Grids Need to Predict Failure — Not React to It



AGING INFRASTRUCTURE

Transformers and grid equipment degrade over time, increasing the probability of failure.



FRAGMENTED SIGNALS

Operators must correlate equipment telemetry, oil diagnostics, and weather conditions to understand risk.



EXTREME WEATHER

Storms, lightning, high winds, and extreme heat can amplify an already degraded asset's risk.

THE CHALLENGE

Identify the assets most likely to fail, understand why, and act before an outage occurs.

Asset degradation + Extreme weather

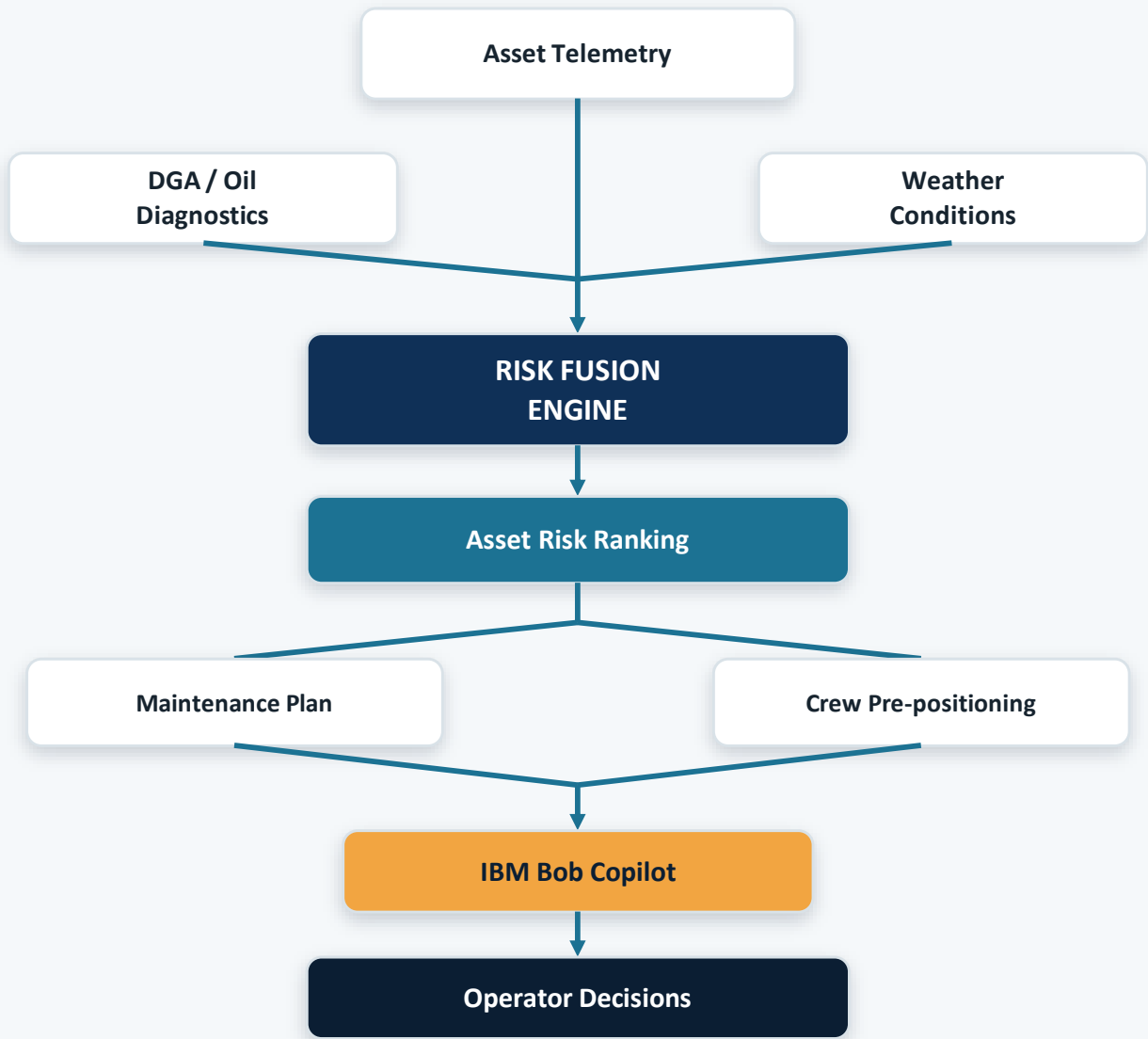


Higher failure risk



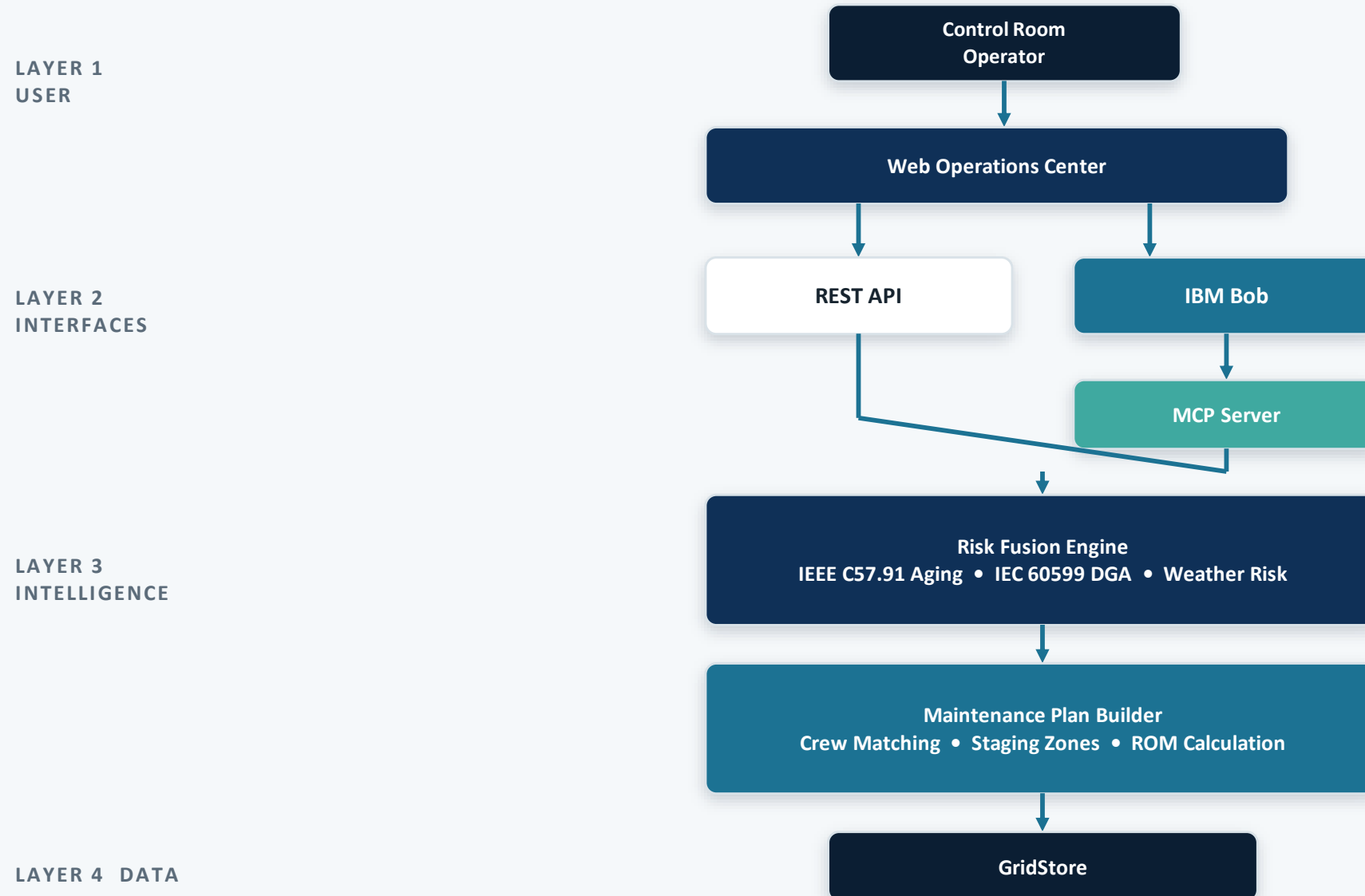
Potential outage

From Raw Grid Telemetry to Preventive Action



GridGuard AI combines physics-based asset diagnostics, weather risk, and operational planning to turn grid data into actionable decisions.

Architecture — Physics + Operations + IBM Bob



Operator-Centric Grid Risk & Scenario Analysis



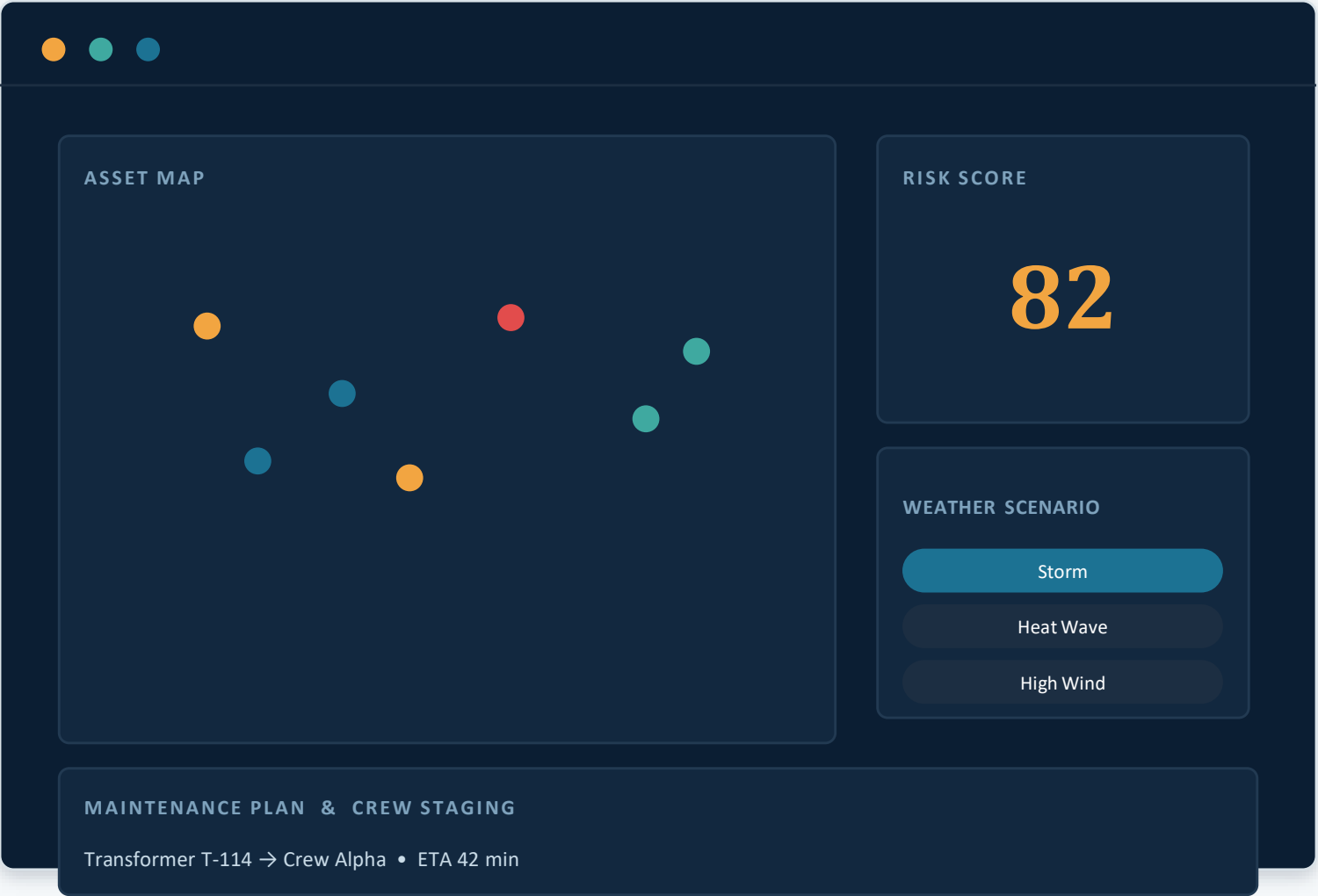
Asset Risk

Identify and rank high-risk grid assets



Weather Scenario

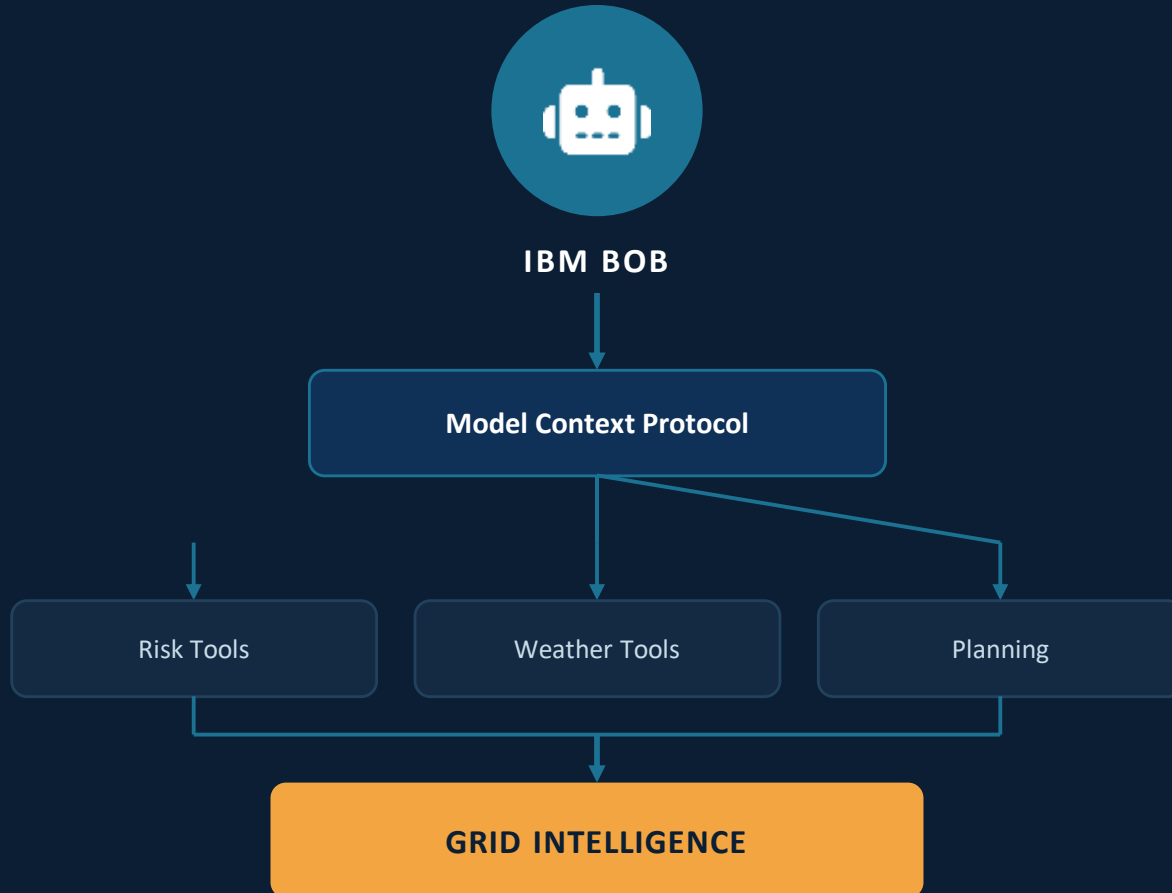
Simulate how severe weather compounds asset risk



Crew Planning

Generate prioritized maintenance and crew staging plans

IBM Bob — The Grid Intelligence Copilot



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MCP TOOLS

Risk assessment, asset ranking, maintenance planning, scenario simulation and more.

4

MCP RESOURCES

Live grid KPIs, assets, weather, and latest plans.

3

OPERATIONAL WORKFLOWS

Pre-storm briefing • Asset deep dive • Shift handoff report

Bob is not just used to build the project — it is integrated as an operational AI interface to the grid intelligence backend.

Turning Prediction Into Grid Resilience

01



EARLIER DETECTION

Identify high-risk equipment before failure.

02



PRIORITIZED MAINTENANCE

Focus limited maintenance resources on the assets with the highest operational risk.

03



PRE-STORM READINESS

Pre-position crews and prepare contingency actions before severe weather arrives.

04



PROTECTED CRITICAL LOADS

Support contingency planning for critical customers such as hospitals and industrial facilities.

Predict → Prioritize → Prepare → Protect

Built by GridGuard AI



Team Member 1

Role



Team Member 2

Role



Team Member 3

Role



Team Member 4

Role

OUR VISION

A resilient grid should know where failure is coming from — and act before customers feel it.